Assignment -16

(Interface)

1. Define an interface “IntOperations” with methods to check whether a number is positive/ negative, even/odd, prime, palindrome and operations like factorial and sum of digits. Define a class MyNumber having one private data member of type int. Write a default constructor to initialize it to 0 and another constructor to initialize it to a value (Use this). Implement the above interface. Create an object in main method. Input a number and write a menu driven program to check different properties of the number using above methods.

2. Define an interface “StackOperations” which declares methods for a static stack. Define a class “MyStack” which contains an array and top as data members and implements the above interface. Initialize the stack using a constructor. Write a menu driven program to perform all operations(Push, POP, Peak) on a MyStack object.